1. Introduction

Access to services and facilities is uneven across different parts of Greater Sydney. Areas with higher concentrations of restaurants, transport services, healthcare facilities, and commercial activity are often considered to be better resourced and more accessible than areas with fewer available services. Understanding these spatial differences is important for examining patterns of urban development, accessibility, and regional inequality within metropolitan Sydney.

This project investigates the spatial distribution of Points of Interest (POIs) across selected regions of Greater Sydney using spatial analysis and database techniques. POI data were collected through the NSW Points of Interest API and combined with ABS SA2 boundary data to analyse the distribution of services and facilities at different geographic scales.

To support the analysis, POI records were processed using GeoPandas, PostgreSQL, and PostGIS. Spatial joins were used to assign POIs to SA2 regions, allowing comparisons of POI concentration and distribution across selected SA4 regions. The project also incorporated regional summary indicators and derived statistics to provide broader demographic and economic context for the spatial analysis.

By combining spatial boundary data, API-based POI extraction, and spatial database methods, this study demonstrates how geographic data technologies can be used to explore patterns of urban accessibility and resource distribution across Greater Sydney

2. Data and Methodology

2.1 Study Area and Spatial Boundary Data

This study focused on selected subregions within Greater Sydney using the 2021 ABS SA2 boundary shapefile (EPSG:4283, GDA94) as the main spatial framework. The shapefile was imported into Python using GeoPandas, and 373 SA2 regions classified under the "Greater Sydney" GCC were filtered from the national dataset.To balance computational efficiency with coverage of different urban characteristics, four SA4 regions were selected for detailed analysis: Sydney – Parramatta, Sydney – Inner South West, Sydney – City and Inner South, and Sydney – North Sydney and Hornsby. These regions contained 128 SA2 units in total.

2.2 POI Data Collection and Processing

POI data were collected using the NSW Points of Interest API. Because the API accepts rectangular query extents rather than polygon geometries, bounding boxes were generated from SA2 polygon coordinates. All requests used EPSG:4283 to maintain consistency with the ABS boundary dataset.The initial extraction returned 18,209 raw POI records containing attributes such as POI name, category, coordinates, urbanity classification, and temporal metadata. POIs were treated as indicators of urban resources and accessibility across Greater Sydney.Before analysis, records without valid coordinates were removed, duplicate POIs were eliminated using object identifiers, and Unix millisecond timestamps were converted into standard date formats. After cleaning, 10,627 POI records remained. Longitude and latitude coordinates were converted into spatial point geometries using GeoPandas, and the dataset was transformed into a GeoDataFrame using EPSG:4283.Because bounding box queries may include POIs outside SA2 polygon boundaries, a spatial join using the ST_Within function in PostGIS was applied to assign each POI to its containing SA2 region. POIs outside the selected study regions were removed, leaving 9,042 valid POI records.

2.3 Spatial Database and Analysis

A PostgreSQL database with the PostGIS extension was used to store and process spatial data within a dedicated schema named sydney_analysis. SA2 regions were stored as MULTIPOLYGON geometries and POIs as point geometries, both using SRID 4283. The database structure supported efficient spatial querying, joins, and aggregation operations. SQL aggregation queries were used to calculate POI counts at both SA2 and SA4 levels. These aggregated results were then used to compare POI density, spatial distribution, and patterns of urban resource accessibility across Greater Sydney.

3.Results and Findings

3.1 Findings from Derived Statistics

Task 1 provides the NSW-level background for the subsequent POI-based well-resourced score analysis through five derived statistics. These five indicators respectively focus on population growth, gender ratio, natural growth, business density and housing market activity. These indicators are all selected from the data with fewer missing values. Judging from the results, NSW has maintained population growth in most years. Although the population growth rate slightly declined in 2021, it began to rise again afterwards. The growth rate is approximately 2.14% in 2023 and 1.66% in 2024. The Female-to-male population ratio has been slightly higher than 1 throughout the period, indicating that the number of Female population has always been slightly larger than that of male population. In terms of natural growth, births minus deaths decreased from 44,807 in 2021 to 32,496 in 2023. This indicates that the contribution of natural population growth has weakened. In contrast, the business density has increased. The number of businesses per 1,000 people rose from 96.94 in 2020 to 105.73 in 2024, indicating an increase in business activities relative to the population size. The fluctuations in housing market activity were more pronounced. house transfers per 1,000 people reached 15.58 in 2021, then dropped to 10.34 in 2023 and slightly rebounded to 11.35 in 2024.

Overall, these derived statistics show that NSW simultaneously experiences population growth, increased commercial density, weakened natural growth, and fluctuations in housing transaction activity. These findings provide the background for the subsequent analysis of the well-resourced score, as POI-based resource availability may be affected by changes in population, economic activities and the housing market.

3.2 Well-resourced Score Distribution

Figure 1 shows the distribution of well-resourced scores in each SA2 region in this analysis. Overall, the scores of most SA2 regions are concentrated between 0.35 and 0.55, indicating that the POI-based resource availability in most regions is at a medium level rather than particularly high. However, this distribution is not completely uniform. It can be seen from the figure that the scores of a few SA2 regions are significantly higher, especially in regions above 0.8, some of which are even close to 1.0. This indicates that the POI density in these areas is significantly higher than that of most SA2. Therefore, it is regarded as more well-resourced in the scoring. Overall, this histogram indicates that there are differences in the distribution of resources among different SA2 regions. Most areas are at a medium level, but a few areas have a significantly higher concentration of resources. POI-based resources are not evenly distributed across all SA2 regions.

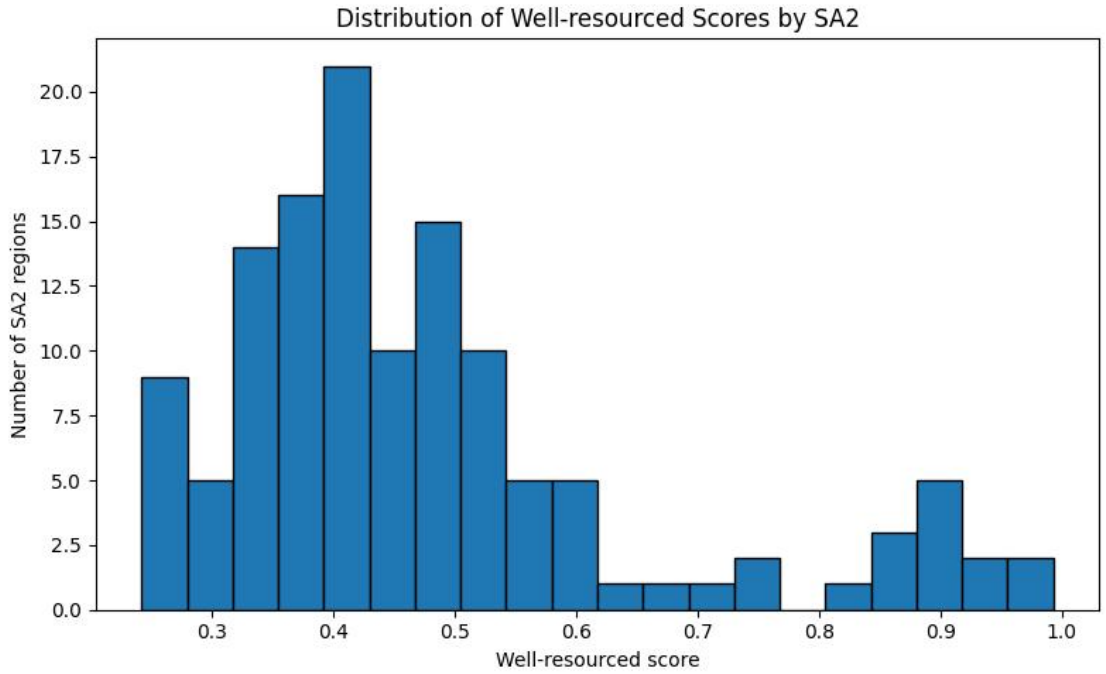


Figure 1. Distribution of Well-resourced Scores by SA2

Figure 2 further shows the overall distribution range of the well-resourced score. It can be seen that the median is approximately 0.43. This indicates that the resource scores of most SA2 regions are not particularly high, but are concentrated at a moderately low to medium level. The boxes are mainly distributed between approximately 0.35 and 0.55, which also indicates that the POI-based resource availability in many SA2 regions is relatively close. However, some obvious high score outliers also appeared on the right side of the figure, especially in the area above 0.8. This indicates that the resource scores of a few SA2 regions are significantly higher than those of other regions. It might be because the POI density in these places is higher, and services, businesses or public facilities are more concentrated. This result is consistent with the previous histogram. Most areas are at a medium level, but a few areas are significantly more well-resourced. Therefore, the overall score distribution is not completely balanced.

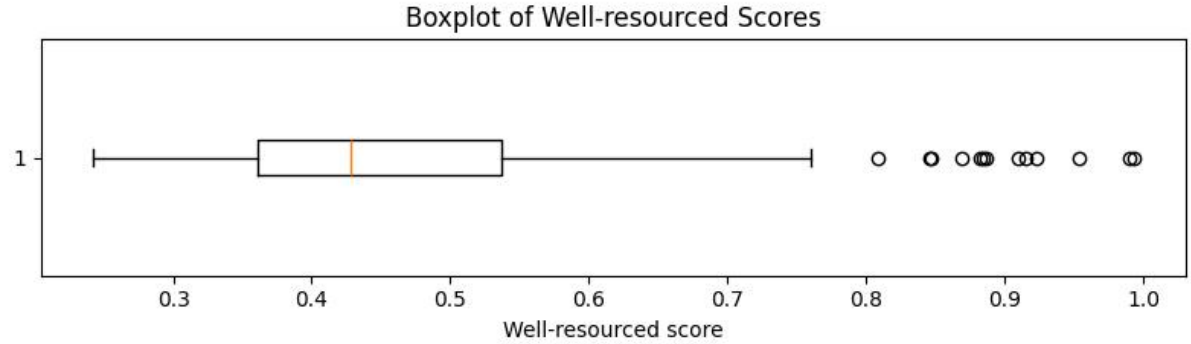


Figure 2. Boxplot of Well-resourced Scores

Table 1 shows the top 10 SA2 regions with the highest well-resourced scores. Most of the top-ranked regions are located in areas that are more central or have a higher concentration of service facilities. For example, Sydney (North) - Millers Point, Sydenham - Tempe - St Peters, Arncliffe - Bardwell Valley and Newtown (NSW). The POI density per square kilometre in these regions is significantly higher, and therefore the final well-resourced score is also higher. Among them, the POI density of Sydney (North) - Millers Point reached 98.37 per sqkm, and the well-resourced score was close to 0.99, which was one of the highest regions among all SA2. Similarly, Sydenham-Tempe-St Peters also has a relatively high POI count and POI density. Therefore, the score is close to 0.99. This indicates that under this scoring model, regions with smaller areas but concentrated POIs are more likely to achieve higher scores. Overall, the Top 10 regions generally have a relatively high POI density, rather than simply scoring high due to a large number of POI counts. This also indicates that the well-resourced score places more emphasis on the concentration degree of resources within a unit area. Therefore, these high-score regions can be understood as areas with denser POI-based resources and stronger service accessibility.

	rank	sa4_name21	sa2_name21	poi_count	area_sqkm	poi_density_per_sqkm	well_resourced_score
0	1	Sydney - City and Inner South	Sydney (North) - Millers Point	316	3.2122	98.374946	0.993186
1	2	Sydney - City and Inner South	Sydenham - Tempe - St Peters	354	3.8539	91.855004	0.989767
2	3	Sydney - Inner South West	Arncliffe - Bardwell Valley	290	4.3130	67.238581	0.953619
3	4	Sydney - City and Inner South	Newtown (NSW)	92	1.5707	58.572611	0.922603
4	5	Sydney - North Sydney and Hornsby	North Sydney - Lavender Bay	109	1.9142	56.942848	0.914958
5	6	Sydney - City and Inner South	Darlinghurst	48	0.8569	56.015871	0.910310
6	7	Sydney - City and Inner South	Sydney (South) - Haymarket	56	1.0810	51.803885	0.886198
7	8	Sydney - Inner South West	Sans Souci - Ramsgate	238	4.6115	51.610105	0.884963
8	9	Sydney - Inner South West	Kingsgrove (South) - Bardwell Park	194	3.7871	51.226532	0.882484
9	10	Sydney - City and Inner South	Potts Point - Woolloomooloo	72	1.4636	49.193769	0.868561

Table 1. Top 10 SA2 Regions by Well-resourced Score

Table 2 shows the 10 SA2 regions with the lowest well-resourced scores. The POI density per square kilometre in these regions is significantly lower than that in the high-score regions. For instance, Banksmeadow has only 4 POIs, and the POI density is approximately 1.02 per sqkm. Therefore, it scored the lowest, approximately 0.24. Yennora Industrial also only had 3 POIs, with a score of approximately 0.25. Many of these low-score areas are Industrial zones or special functional areas, such as Port Botany Industrial, Smithfield Industrial, Yennora Industrial and Sydney Airport. Therefore, a low score does not necessarily indicate that the residents' services in these areas are poor. It could also be because these regions themselves are not primarily for living or providing daily public services. Overall, Table 2 indicates that a lower well-resourced score is usually associated with a lower POI density, but the differences in land use among different Sa2 regions also need to be considered when interpreting these results.

	rank	sa4_name21	sa2_name21	poi_count	area_sqkm	poi_density_per_sqkm	well_resourced_score
118	119	Sydney - North Sydney and Hornsby	Turramurra	124	20.6426	6.006995	0.304010
119	120	Sydney - City and Inner South	Port Botany Industrial	11	2.8044	3.922408	0.276998
120	121	Sydney - North Sydney and Hornsby	Asquith - Mount Colah	120	34.6889	3.459320	0.271202
121	122	Sydney - Inner South West	Chullora	13	4.0501	3.209797	0.268111
122	123	Sydney - Parramatta	Rookwood Cemetery	9	3.0150	2.985075	0.265346
123	124	Sydney - Parramatta	Smithfield Industrial	11	4.3017	2.557129	0.260132
124	125	Sydney - North Sydney and Hornsby	Berowra - Brooklyn - Cowan	178	81.7548	2.177242	0.255559
125	126	Sydney - City and Inner South	Sydney Airport	13	8.9106	1.458937	0.247059
126	127	Sydney - Parramatta	Yennora Industrial	3	2.0787	1.443210	0.246875
127	128	Sydney - City and Inner South	Banksmeadow	4	3.9384	1.015641	0.241908

Table 2. Bottom 10 SA2 Regions by Well-resourced Score

3.3 Spatial Visualisation and Regional Patterns

Figure 3 shows the spatial distribution of well-resourced scores in different SA2 regions on the map. The brighter areas in the figure represent higher scores, while the darker areas represent lower scores. Overall, the high-score areas are not evenly distributed but mainly concentrated in some more central regions with denser service facilities.

At the same time, some darker low-score areas can also be seen on the map. These areas may be located in relatively peripheral positions or belong to special functional zones such as industrial areas, airports, and cemeteries. Therefore, the POI density is relatively low, and the final score is also relatively low. This indicates that the well-resourced score is largely influenced by the degree of spatial concentration of POI. Overall, the results of this map are consistent with those of the previous histogram, box plot, and Top/Bottom SA2 table. A few SA2 regions have a higher POI-based resource availability, while the resource concentration in other regions is relatively low. POI-based resources are not evenly distributed in space.

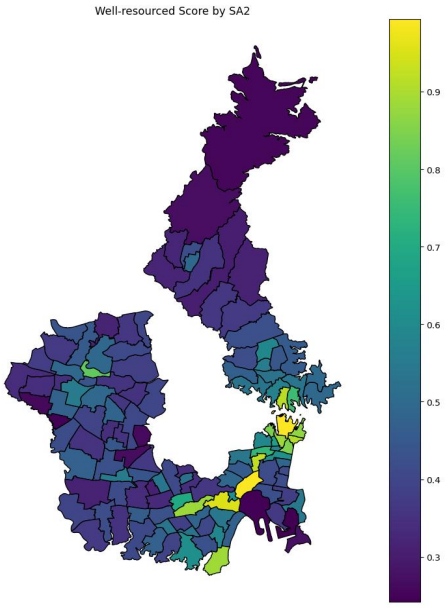


Figure 3. Spatial Distribution of Well-resourced Scores by SA2

4. Discussion

4.1 Uneven Distribution of Urban Resources

Looking at the results across the chosen Greater Sydney SA4s, it is clear that POI resources are not spread evenly. High scores are heavily clustered in central and inner-city SA2s like Millers Point and Newtown. This makes sense, as these areas are established commercial hubs with major transport links. Interestingly, even though our derived statistics showed a general rise in business density across NSW, the spatial map proves this growth is mostly concentrated in a few specific urban pockets rather than benefiting the broader suburban areas.

4.2 The Impact of Land Use Zoning

Land use types also heavily skew the well-resourced scores. The lowest-ranking SA2s are almost entirely industrial zones (like Port Botany and Yennora) or transport infrastructure (Sydney Airport). A low score here doesn't mean residents are missing out on services; it simply reflects the fact that these areas are not meant for residential living. This highlights a flaw in blindly using POI density as a measure of livability; we always need to compare the data with actual land zoning.

4.3 Urban Planning Takeaways

Most SA2s scored somewhere in the middle (0.35 to 0.55), meaning the average suburb isn't particularly resource-rich. Given NSW's growing population and shifting housing market, these findings suggest that future urban planning needs to focus more on decentralizing services. Building up local facilities in the intermediate suburbs and outer suburbs would help reduce the heavy reliance on the inner city and improve overall accessibility for residents.

5. Limitations

There are a few limitations in our approach that should be noted:

5.1 API Data Constraints

A major issue with relying on the NSW POI API is that every facility is counted as a single point, ignoring its actual size or capacity. For example, a massive public hospital and a small local GP clinic both contribute exactly '1' to the POI count, even though the hospital serves far more people. The API dataset might also be slightly outdated, missing newer businesses or containing coordinate errors.

5.2 Spatial Boundaries and MAUP

Using SA2 boundaries introduces spatial biases, commonly known as the Modifiable Areal Unit Problem (MAUP). Since our score depends on POI density (POIs per square kilometre),

SA2s that include large areas of bushland or national parks automatically get lower scores, even if their residential streets are packed with shops. Also, using a strict spatial join (ST_Within) creates artificial borders. A resident living right on the edge of a low-score SA2 might just cross the street to use services in a high-score SA2, but our current geographic boundaries fail to capture this real world mobility.

5.3 Ignoring Population Demand and Transport

Our current scoring model only measures how tightly packed the POIs are, without accounting for population demand or actual travel times. A dense cluster of services isn't useful if there is no public transport to get there. To make this analysis more robust in the future, we would need to weight the POI counts against local population data and map out actual walking or driving distance catchments, rather than just relying on polygon densities.

6. Conclusion

This study used POI-based spatial analysis to examine patterns of urban accessibility across selected regions of Greater Sydney. The results showed that services and facilities were concentrated mainly in highly developed inner city areas, while many suburban, industrial, and areas primarily serving transport functions recorded lower well-resourced scores. This suggests that resource accessibility across Greater Sydney is spatially uneven rather than evenly distributed.

By combining GeoPandas, PostgreSQL, and PostGIS, the project demonstrated how GIS and spatial database techniques can effectively support urban spatial analysis and reveal regional accessibility disparities. Although the analysis was constrained by API data limitations and the use of SA2-based spatial boundaries, the workflow provides a useful foundation for future studies of urban planning, service accessibility, and resource distribution.